

Goal

Estimate how long `code-A.R` may take and how to check it safely before running the full MANOVA.

Problem

`code-A.R` fits a MANOVA with three response variables:

```
cbind(rmse, L1, Linf)
```

and a relatively large design matrix with interactions between `method` and several design factors, plus `p1` and `p2`.

The runtime depends mostly on:

```
nrow(acc_mv)
number of factor levels
available RAM
CPU speed
whether R is 64-bit
```

Practical estimate

For your 972 experiments, I would expect roughly:

| Size of <code>acc_mv</code> | Likely runtime for <code>code-A.R</code> |
|-----------------------------|------------------------------------------|
| < 100,000 rows              | seconds to 1 minute                      |
| 100,000–500,000 rows        | 1–5 minutes                              |
| 500,000–2,000,000 rows      | 5–30 minutes                             |
| > 2,000,000 rows            | 30+ minutes, possible memory limits      |

The MANOVA itself is usually faster than the mixed model in Analysis B, but it can still be memory-heavy because `manova()` builds a model matrix internally.

Exact R syntax to estimate runtime before running the full analysis

After loading `acc_mv`, run:

```
nrow(acc_mv)
object.size(acc_mv)
format(object.size(acc_mv), units = "GB")
```

Then check the size of the model matrix:

```
X <- model.matrix(
  ~ method * design_block +
  method * G +
  method * theta_true +
  method * theta_CV +
  method * sigma +
  method * mean_nsamp +
  method * p1 + method * p2,
  data = acc_mv
)

dim(X)
format(object.size(X), units = "GB")

rm(X)
gc()
```

This tells you the real memory pressure before fitting MANOVA.

## Time a smaller test run

Use a 5% sample first:

```
set.seed(123)

acc_mv_test <- acc_mv[sample(seq_len(nrow(acc_mv)),
  size = ceiling(0.05 * nrow(acc_mv))), ]

system.time({
  fit_manova_test <- manova(
    cbind(rmse, L1, Linf) ~ method * design_block +
    method * G +
    method * theta_true +
    method * theta_CV +
    method * sigma +
    method * mean_nsamp +
    method * p1 + method * p2,
    data = acc_mv_test
  )
})
```

If the 5% run takes 30 seconds, the full run may take more than 10 minutes because model fitting often scales worse than linearly with the number of rows and model columns.

## Full run with timing

```
gc()

system.time({
  fit_manova <- manova(
    cbind(rmse, L1, Linf) ~ method * design_block +
      method * G +
      method * theta_true +
      method * theta_CV +
      method * sigma +
      method * mean_nsamp +
      method * p1 + method * p2,
    data = acc_mv
  )
})

system.time({
  print(summary(fit_manova, test = "Pillai"))
})

system.time({
  print(summary.aov(fit_manova))
})
```

## Bottom line

If `acc_mv` has fewer than about **500,000 rows**, `code-A.R` may finish in a few minutes. If it has **millions of rows**, it may take tens of minutes and may still hit memory limits. The safest next step is to run the `model.matrix()` size check and the 5% timing test before running the full MANOVA.